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Computer Networking

Lab 05: Configuring Network Devices

A. OVERVIEW

1. Objectives

- Students will be able to configure wireless network devices.
- Students will apply IP address subdivision within a network model.

2. General Knowledge

- Knowledge of network devices and IP addressing.

3. Environment & Practical Tools

- A computer with Internet connectivity running the Windows operating system.
- **Packet Tracer Software:** Students can download the latest free version following the instructions at https://www.netacad.com/courses/packet-tracer

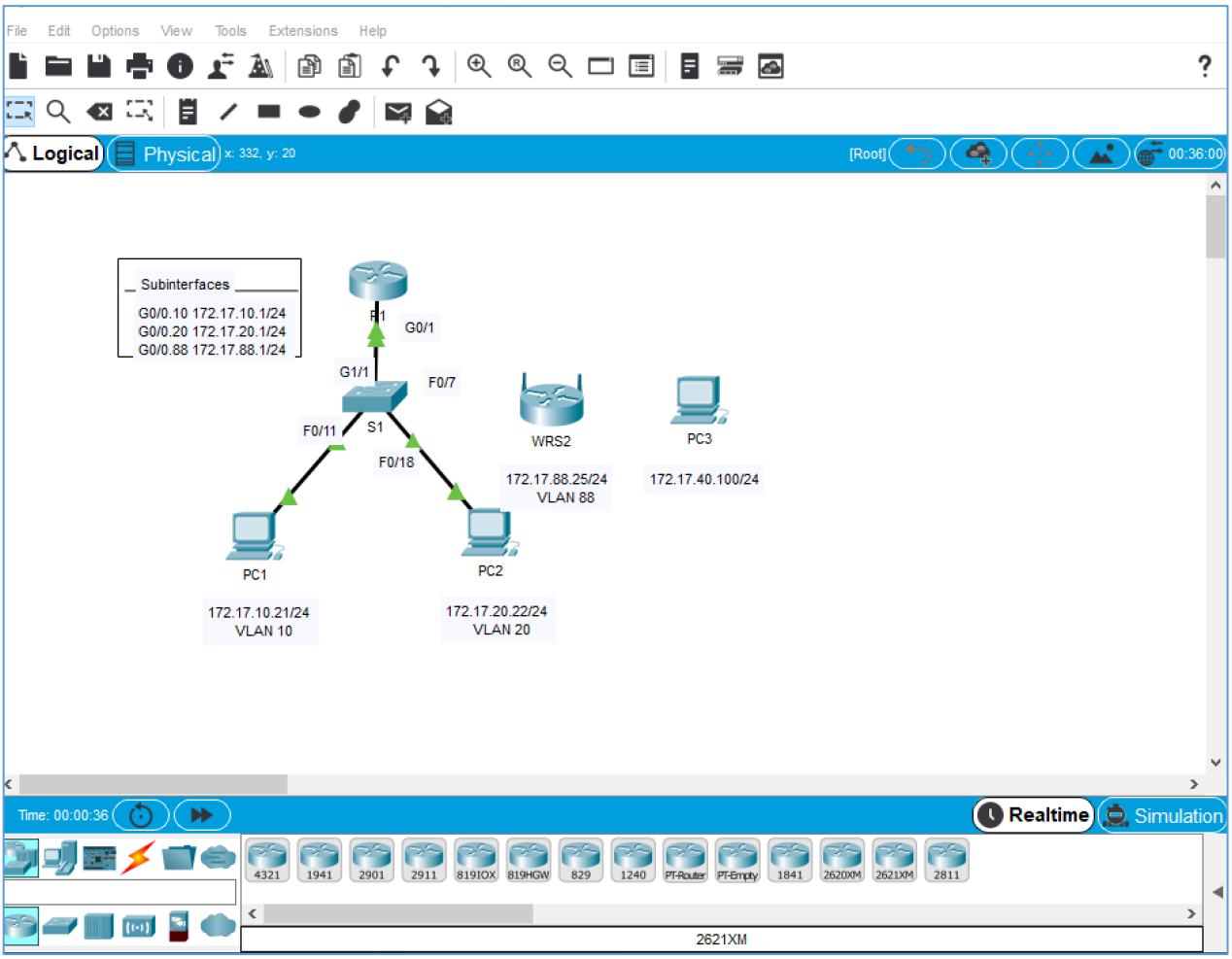

Fig 1. Main Interface of the Packet Tracer Software.

B. PRACTICAL EXERCISES

1. Task 1: Configuring Wireless Network Devices

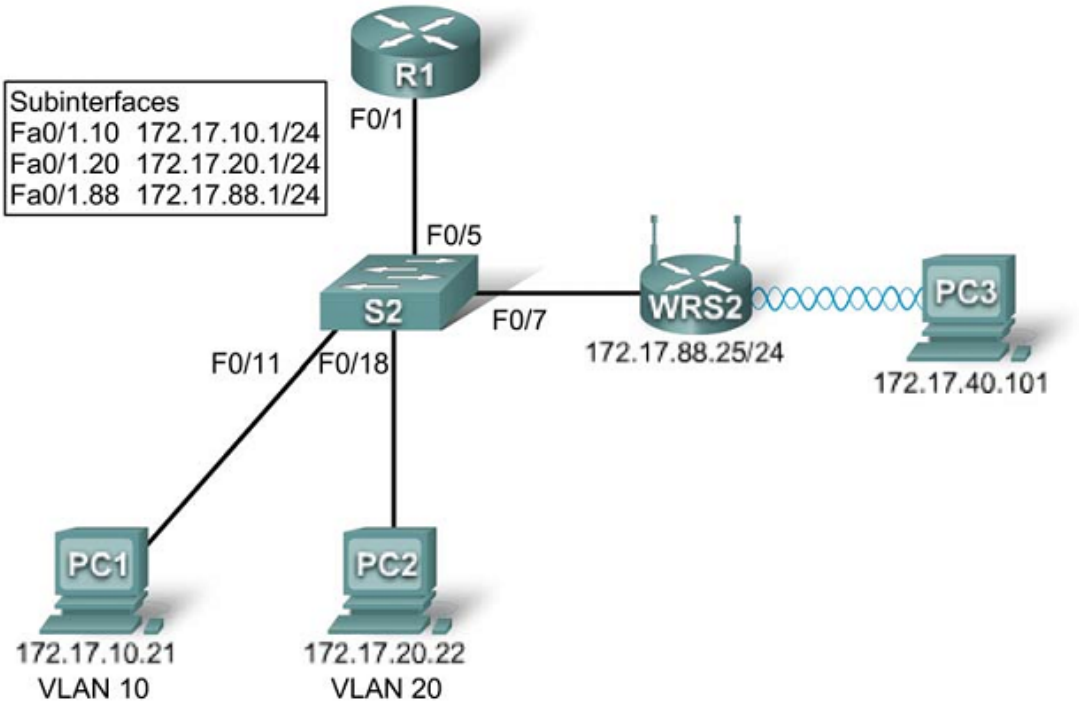

Fig 2. Network Model for Wireless Network Device Configuration Practice.

Open the **Lab 5a.pka** file and follow the instructions below:

1.1 Connecting the Wireless Network Device to the Model

Use a **Copper Straight-through** cable to connect the Internet port of the wireless router to the Fa0/7 port of the switch.

1.2 Basic Configuration

- **Configuring the Internet Connection Section**

Click **WRS2 > GUI** tab

Set the **Internet Connection type** to **Static IP**.

Configure the IP address section with the following parameters:

- Internet IP address: 172.17.88.25.
- Subnet mask: 255.255.255.0.
- Default gateway: 172.17.88.1

- **Configuring the Network Setup Section**

Scroll down to the **Network Setup**. Enter the **Router IP** as **172.17.40.1** and the subnet mask as **255.255.255.0**.

Select **Enabled** for the DHCP server section.

Click **Save Settings**.

1.3 Configuring Access and Security

Select **Wireless** tab at the top. Change the **Network Mode** to **Wireless-N Only** and change the name of the wireless network (SSID) to **WRS_LAN**.

Disable **SSID Broadcast** (select **Disabled**) and click **Save Settings**.

Select the **Wireless Security** tab. Change the **Security Mode** from **Disabled** to **WPA2 Personal**.

WEP, WPA, WPA2,…

Use the password **cisco123** (in the passphrase field)

Click **Save Settings**.

1.4 Configuring the Wireless Client

Since SSID broadcast is disabled, the wireless settings for PC3 must be manually configured.

Click **PC3** > **Desktop** > **PC Wireless**.

Click **Profiles** tab.

Click **New**.

Name the profile **Wireless Access**.

In the next screen, click **Advanced Setup**. Type **WRS_LAN** in the **Wireless Network Name** field. Click **Next**.

Select **Obtain network settings automatically (DHCP)** and click **Next**.

In the **Wireless Security** section, select **WPA2-Personal** and click **Next**.

Type the passphrase as **cisco123** and click **Next**.

Click **Save** and click **Connect to Network**.

1.5 Connection Verification

View the **Signal Strength** and **Link Quality** sections for connection link information.

Click **More Information** to view the detailed IP address configuration.

1.6 Verifying Practical Results

Upon completion, students should switch to the PT Activity window and press Check Results to view the outcome.

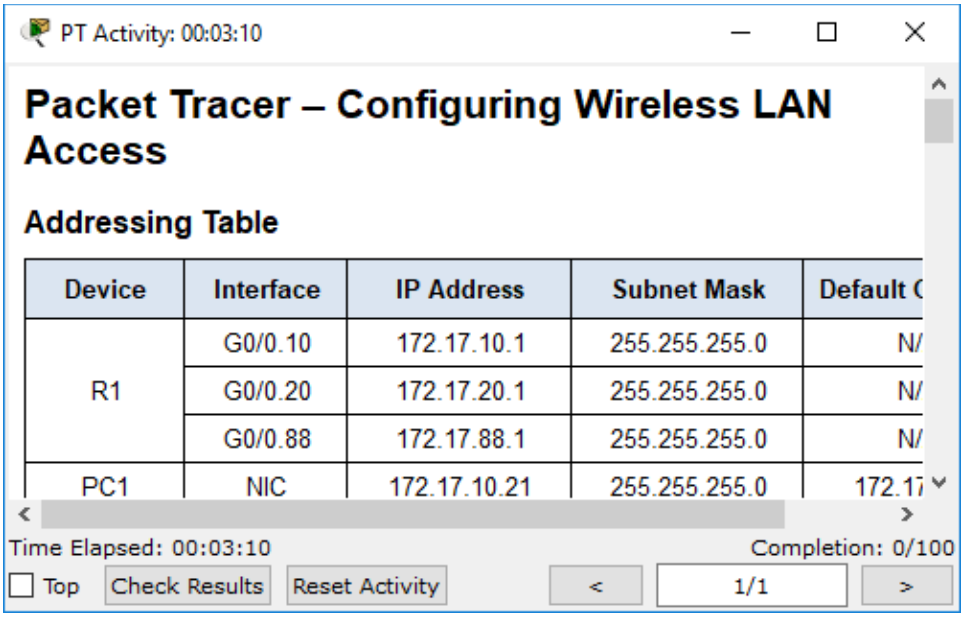

Fig 3. PT Activity Window.

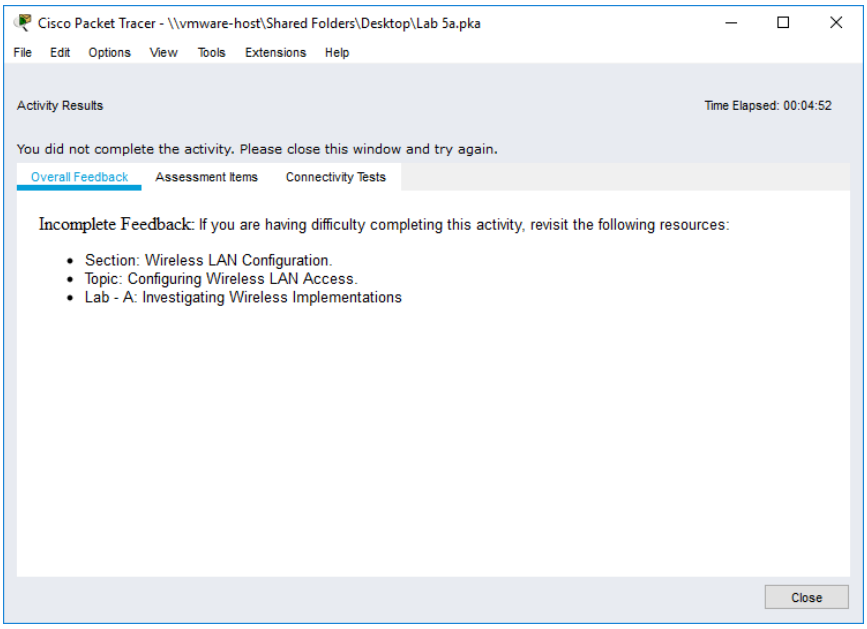

Fig 4. Overall Results before Configuration Implementation.

Select the Assessment Items tab to view detailed results.

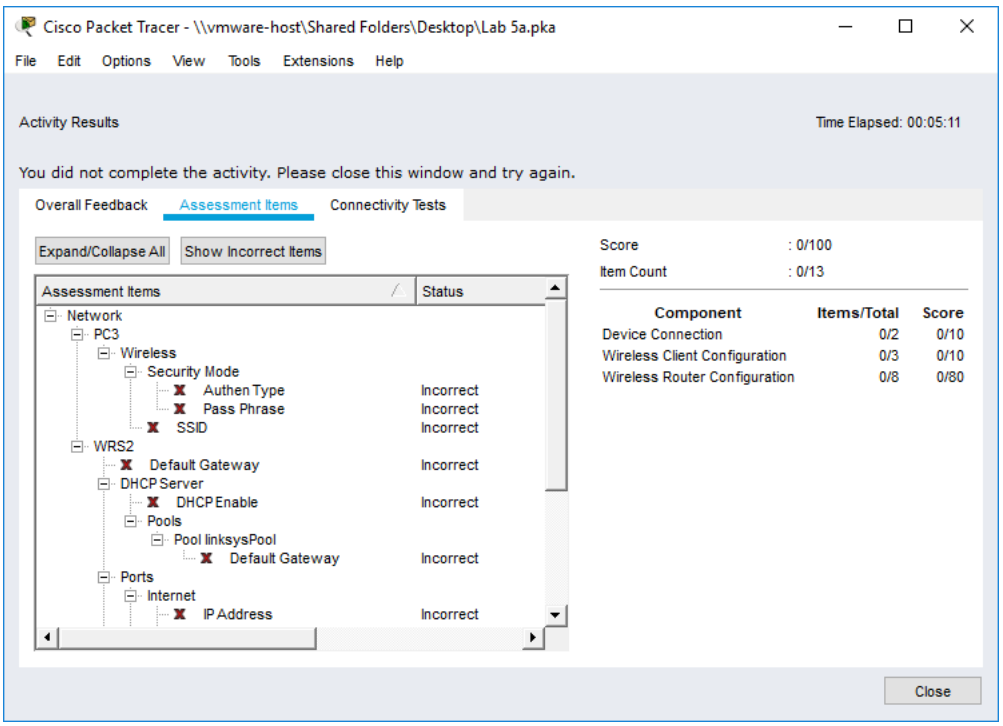

Fig 5. Detailed Results before Configuration Implementation.

Requirement: *Capture images of the test results (overall and detailed) for inclusion in the report. Save the file with the name Lab 5a_MSSV.pka*

2. Task 2: Configuring IP Addresses on a Router

Open the *Lab 5b.pka* file and follow the instructions below:

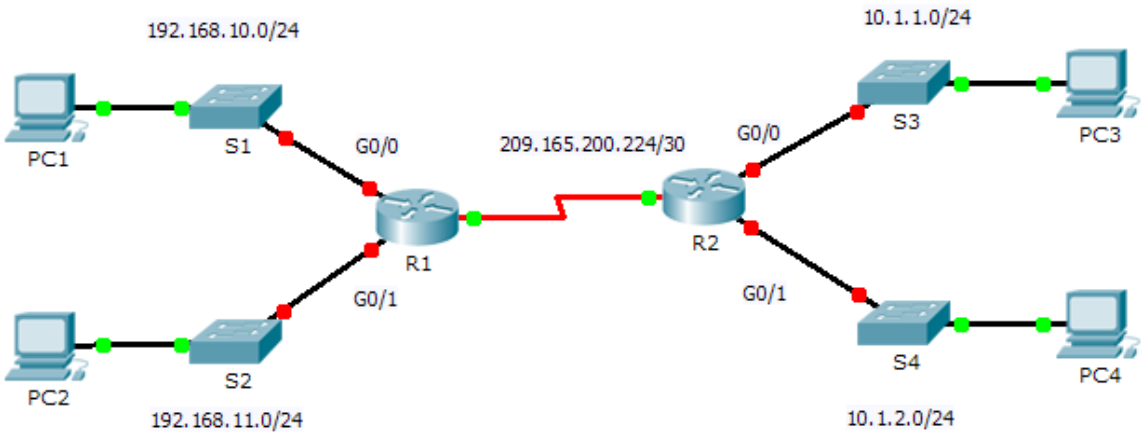

Fig 6. Network Model for IP Address Configuration Practice on a Router.

IP Address Table:

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0	192.168.10.1	255.255.255.0	N/A
	G0/1	192.168.11.1	255.255.255.0	N/A
	S0/0/0 (DCE)	209.165.200.225	255.255.255.252	N/A
R2	G0/0	10.1.1.1	255.255.255.0	N/A
	G0/1	10.1.2.1	255.255.255.0	N/A
	S0/0/0	209.165.200.226	255.255.255.252	N/A
PC1	NIC	192.168.10.10	255.255.255.0	192.168.10.1
PC2	NIC	192.168.11.10	255.255.255.0	192.168.11.1
PC3	NIC	10.1.1.10	255.255.255.0	10.1.1.1
PC4	NIC	10.1.2.10	255.255.255.0	10.1.2.1

2.1 Configuring the IP Address for Router R1

To configure the IP address for the router Interfaces, use the following commands:

First, enter configuration mode (`R1(config)#`)

```
R1>enable
R1#config t
R1(config)#
```

The passwords are: **cisco** and **class**, respectively.

To configure the address for the gigabitethernet 0/0 interface, use the following commands (the address configured is the one specified in the address table):

```
R1(config)# interface gigabitethernet 0/0
R1(config-if)# ip address 192.168.10.1 255.255.255.0
R1(config-if)# no shutdown
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
```

Configure the addresses for the remaining interfaces: Gi0/1 of R1, and Gi0/0, Gi0/1 of R2, using the addresses from the address table. After completing the configuration, save the configuration using the command:

```
R1#copy run start
```

2.2 Configuration Verification

Verify connectivity by pinging from PC1 -> PC4, R2 ->PC2

Use the **show** commands to view the device configurations (capture images of the output of the following **show** commands):

```
R1#show run
R1#show ip interface brief
R1#show ip route
```

2.3 Verifying Practical Results

Perform similarly to section 1.6.

Requirement: *Capture images of the test results (overall and detailed) for inclusion in the report. Save the file with the name Lab 5b_MSSV.pka*

3. Task 3: Applying IP Address Subdivision

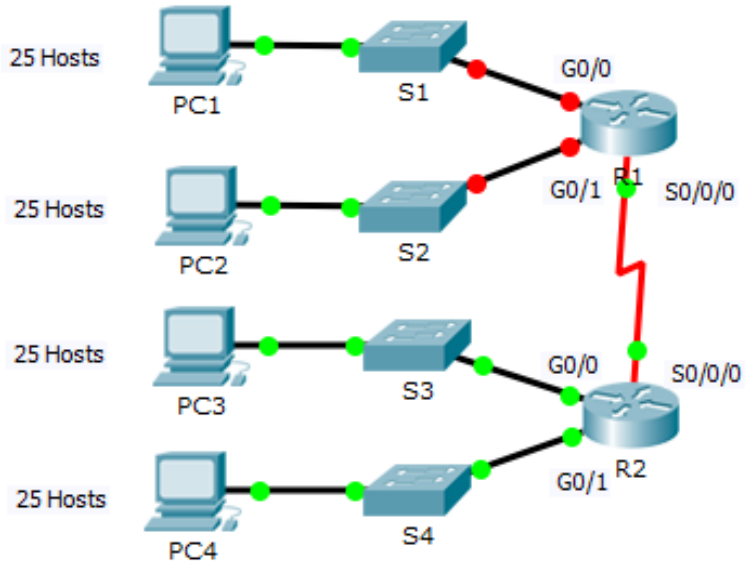

Fig 7. Network Model for IP Address Subdivision Practice.

3.1 Subnetting and IP Allocation

Given the network model above and the address **192.168.100.0/24**, determine the following:

- How many subnets are required?
- Perform subnetting based on the given address. List the parameters including the network address, the host address range, and the broadcast address for those subnets, and complete the table below:

No.	Network Address	Starting Host Address	Ending Host Address	Broadcast Address
0				
1				
2				
3				
4				
5				
6				
7				

After performing subnetting, assign the subnets to each network in the model as follows:

- Assign Subnet 0 to the LAN connected to the GigabitEthernet 0/0 interface of R1
- Assign Subnet 1 to the LAN connected to the GigabitEthernet 0/1 interface of R1
- Assign Subnet 2 to the LAN connected to the GigabitEthernet 0/0 interface of R2
- Assign Subnet 3 to the LAN connected to the GigabitEthernet 0/1 interface of R2
- Assign Subnet 4 to the connection between R1 and R2

After subnet assignment, complete the following address table:

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0			
	G0/1			
	S0/0/0			
R2	G0/0			
	G0/1			
	S0/0/0			
S1	VLAN 1			
S2	VLAN 1			
S3	VLAN 1			
S4	VLAN 1			
PC1	NIC			
PC2	NIC			
PC3	NIC			
PC4	NIC			

Note that:

- Use the first usable address of the subnet for the GigabitEthernet Interfaces of R1 and R2.
- For the connection between R1 and R2, use the first usable address of the subnet for the S0/0/0 Interface of R1 and the last usable address for the S0/0/0 Interface of R2.
- Use the second usable address of the subnet for the Switches.
- Use the last usable address of the subnet for the PCs.

3.2 Device Configuration

Configure the appropriate IP addresses for devices R1, S3, and PC4.

3.3 Verifying Practical Results

Perform similarly to section 1.6.

Requirement: *Capture images of the test results (overall and detailed) for inclusion in the report. Save the file with the name Lab 5c_MSSV.pka*

THE END